Concept Review

Push Buttons

Why Do We Need Push Buttons?

Push buttons are one of the most basic components that collect user inputs—they are found everywhere, ranging from simple light switches to your computer keyboard. As modern engineering systems continue to scale up in complexity, push buttons remain a staple component across a wide range of technology due to their simplicity, reliability, and durability.

Push Button Switches

Push buttons are a type of switch that users manually actuate to open or close an electrical connection. We can read either a high, low, or high impedance state using a push button, depending on the type of push button and how it is connected to the rest of the circuit.

Types of Push Buttons

Push buttons can be classified as Normally Open (NO) or Normally Closed (NC). A **Normally Open** push button means that there is no connection between its terminals (it is "open") in its default state, and actuating the button will complete the connection. A **Normally Closed** switch is closed in its default state and open in its actuated state.

Figure 1: Schematic symbols for Normally Open (left) and Normally Closed (right) push button

Switches can be classified as momentary or latching: a **momentary** switch returns to its default state when it is not actuated, while a **latching** switch retains its current state until the next time it is actuated. By default, push buttons are assumed to be momentary unless it is listed specifically as a latching push button. Latching behavior can be implemented in software by using momentary push button inputs to toggle the value of a state.

Reading Push Button Inputs

To read the input from a pushbutton, it is connected to a digital input pin on a microcontroller, and the other side can be connected either to ground or to a voltage supply.

When the push button is placed between the input pin and the voltage source, closing the connection pulls the signal to high. A **pull-down** resistor, shown in the circuits in Figure 2, should be placed to connect the input pin to ground, which pulls the signal to low when the connection is open. Without the pull-down resistor, the input pin could read the floating signal as either low or high, causing the system to react unpredictably.

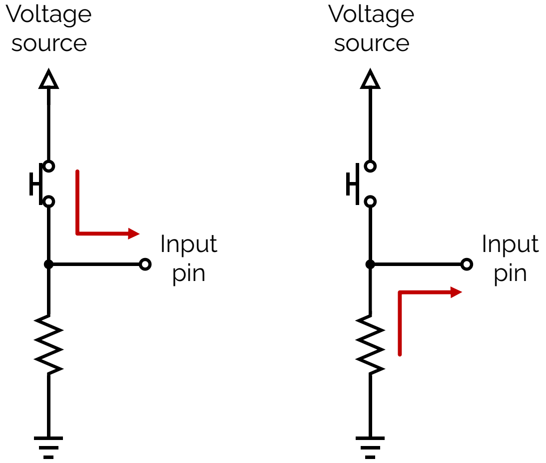

Figure 2: Push button connected with pull-down resistor, in closed and open states

When the push button is wired to ground, closing the connection pulls the signal to low. A **pull-up** resistor, shown in the circuits in Figure 3, should be placed to connect the input pin to the voltage source, which pulls the signal to high when the connection is open. Without the pull-up resistor, the input pin could read either low or high.

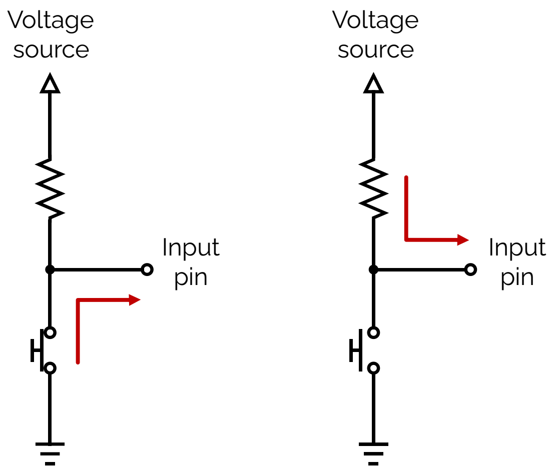

Figure 3: Push button circuits connected with pull-up resistor, in closed and open states

Debouncing Pushbuttons

Since pushbuttons are mechanically actuated, they are susceptible to switch bouncing, which causes the input to fluctuate between high and low before fully settling into the final state when the button is pressed or released.

The effects of switch bouncing can propagate through your system, causing unpredictable or undesired behaviour. To ensure that your system performs reliably, switch debouncing is implemented using either software or hardware techniques to mitigate the effect of switch bouncing. Note that for the Mechatronics Sensors Trainer, debouncing has already been implemented so bouncing behavior is not observed when reading from the pushbuttons on the device.

Datasheet and Considerations

Selecting an appropriate push button involves consideration of how users will interact with the button. Momentary buttons are used when their function is only required when pressed, for example, as keys in a computer keyboard. In cases where the switch needs to hold its state until it is pressed again, such as a light switch, latching switches are more suitable. The amount of force required to change the button state is listed as the **actuation force**. A larger actuation force can prevent accidental triggers at the cost of increasing user fatigue if the user needs to frequently actuate the button.

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